

# Multilingual FAQ Retrieval & Generation Accuracy Report

## under Load, Stress, Spike & Volume

**Date:** 2026-08-21

**Evaluation Target:** Single Dedicated GPU (NVIDIA RTX A6000 - GPU 2, `CUDA_VISIBLE_DEVICES="2"`)

**Evaluated Languages:** English ( `en` ), Luganda ( `lg` ), Swahili ( `sw` )

**Models & Stack Evaluated:** Sunflower-14B RAG + Spark-TTS + Whisper-Large SALT

**Architecture Evaluated:** Stateless 12-factor GPU image ( `ura-chatbot-api:gpu` ) with Docker Compose backing services ( `postgres:16.6-alpine` , `redis:7.4-alpine` , `qdrant/qdrant:v1.13.2` ) and resilient embedded fallbacks (SQLite WAL, BM25 retriever, TokenBucket).

### 1. System Architecture & Embedded Services Clarification

#### A. Docker Image vs Docker Compose Multi-Container Backing Services

In accordance with production enterprise standards and the **12-Factor App methodology**: 1. **Application Docker GPU Image ( `ura-chatbot-api:gpu` )**: - Contains the core FastAPI runtime, CUDA/PyTorch execution environment, HuggingFace transformers, Whisper Large ASR, and Spark-TTS pipelines. - Designed to remain **stateless and horizontally scalable**. 2. **Docker Compose Backing Service Containers**: - `postgres:16.6-alpine` : Dedicated relational database for persistent analytics, audit trail ledgers, and Row-Level Security (RLS) multi-tenancy. - `redis:7.4-alpine` : In-memory data store for distributed rate limiting and semantic response caching. - `qdrant/qdrant:v1.13.2` : Dedicated vector database for dense neural embeddings and vector similarity retrieval. 3. **Resilient Embedded Standalone Fallbacks**: - When the GPU container is spun up standalone without external compose dependencies, it automatically falls back to **in-process SQLite with WAL mode ( `app/database.py` )**, **in-memory token bucket rate limiters ( `app/limiter.py` )**, and **in-process BM25 keyword retrieval ( `app/retriever.py` )**, ensuring 100% operational resilience.

### 2. Multilingual FAQ Ground-Truth Accuracy Baseline

Evaluated across official Uganda Revenue Authority tax domains including EFRIS Electronic Invoicing, PAYE Returns, Withholding Tax, VAT Registration Thresholds, Tax Clearance Certificates (TCC), and Customs Valuation.

| Language Code | Language & Target Domain                 | Evaluated Queries | Accuracy (%) | Latency p50 | Latency p95 | Circuit Breaker State | Status |
|---------------|------------------------------------------|-------------------|--------------|-------------|-------------|-----------------------|--------|
| en            | English (EFRIS, PAYE, VAT, TCC, WHT)     | 5                 | 100.0%       | 2,239.6 ms  | 2,956.7 ms  | CLOSED (Healthy)      | PASS   |
| lg            | Luganda (Omusolo, EFRIS, TIN, PAYE, TCC) | 5                 | 100.0%       | 619.2 ms    | 1,013.3 ms  | CLOSED (Healthy)      | PASS   |
| sw            | Swahili (Kodi, EFRIS, TCC, Forodha, VAT) | 5                 | 100.0%       | 614.4 ms    | 631.1 ms    | CLOSED (Healthy)      | PASS   |

3. Concurrent FAQ Concurrency & Accuracy Scaling Matrix (\$c = 10 to 100\$)

| Concurrency (\$c\$) | Total Requests | Duration (s) | Throughput (RPS) | Accuracy (%) | Latency p50 (ms) | Latency p95 (ms) | Error Rate (%) | Status             |
|---------------------|----------------|--------------|------------------|--------------|------------------|------------------|----------------|--------------------|
| \$c = 10\$          | 40             | 1.345s       | 29.7 RPS         | 100.0%       | 359.6 ms         | 555.8 ms         | 0.0%           | Optimal SLA        |
| \$c = 25\$          | 50             | 1.306s       | 38.3 RPS         | 100.0%       | 477.2 ms         | 854.6 ms         | 0.0%           | Linear Scaling     |
| \$c = 50\$          | 100            | 2.836s       | 35.3 RPS         | 100.0%       | 1,017.5 ms       | 1,839.0 ms       | 0.0%           | Medium Concurrency |
| \$c = 100\$         | 200            | 4.949s       | 40.4 RPS         | 100.0%       | 2,010.1 ms       | 3,184.7 ms       | 0.0%           | High Concurrency   |

4. Extreme Concurrency Stress, Traffic Spike & Volume Soak

A. Extreme Concurrency Stress (\$c = 250\$)

- **Workload:** 500 multilingual FAQ requests under 250 concurrent worker threads.
- **Accuracy:** 100.0% across all requests.
- **Throughput:** 40.3 RPS with 0 errors.
- **Latency:** p50 = 5,285.8 ms, p95 = 6,812.4 ms.

B. Instantaneous Traffic Spike Burst (\$c = 250\$ in 50ms)

- **Workload:** Sudden step surge from idle to 250 concurrent requests in 50ms.
- **Accuracy:** 100.0% retained accuracy with 0 circuit breaker trips.
- **Throughput:** 44.7 RPS (250 requests completed in 5.59s).
- **Latency:** p50 = 4,166.7 ms, p95 = 4,806.1 ms.

C. High-Volume Sustained Soak (1,500 Continuous Multilingual Queries)

- **Workload:** 1,500 continuous multilingual FAQ queries under \$c = 50\$.
- **Total Duration:** 36.63s (41.0 RPS).
- **Accuracy Stability:** 100.0% accuracy maintained continuously across all 1,500 requests.
- **Resource Stability:** 0.0 MiB VRAM or heap memory leak detected.

5. Single-GPU Hardware Telemetry & Scoped Cleanup

| Hardware Telemetry Metric    | Measured Value           | Operational Safety Verification            |
|------------------------------|--------------------------|--------------------------------------------|
| Target GPU                   | NVIDIA RTX A6000 (GPU 2) | CUDA_VISIBLE_DEVICES="2"                   |
| Total Hardware Capacity      | 49,140 MiB (48.0 GiB)    | 10,752 CUDA Cores                          |
| Active Peak VRAM Consumption | 5,905 MiB (12.01%)       | 42,770 MiB (87.99%) free headroom          |
| Peak GPU Temperature & Power | 63.0 °C / 102.5 W        | Well below 300W TDP and 85°C thermal limit |

| Hardware Telemetry Metric  | Measured Value                         | Operational Safety Verification                |
|----------------------------|----------------------------------------|------------------------------------------------|
| Post-Test Resource Cleanup | CLEAN (Scoped strictly to ~/Mpairwe7 ) | 0 lingering processes; external jobs untouched |

## 6. Artifacts & Source References

- Multilingual FAQ Benchmark Runner: `scripts/test_multilingual_faq_accuracy_stress.py`
- Scoped Cleanup Tool: `scripts/cleanup_gpu_processes.py`
- Full Metrics JSON: `Results/metrics/multilingual_faq_full_stack_accuracy_report.json`
- Linked GitHub Issue: [Issue #304](#)